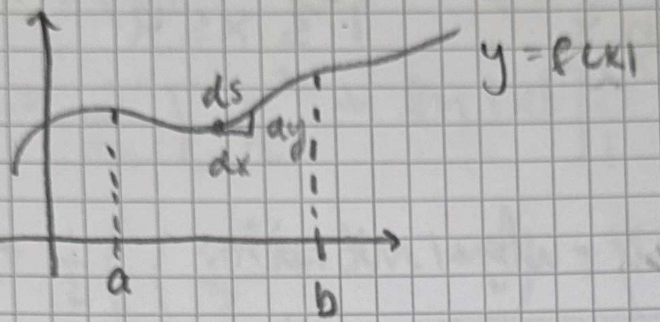
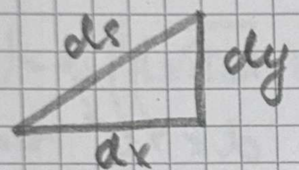
ENVARIABELSANALYS DEL 2
SEMINARIUM 10: kurvlängder



ds i bilden är bågelement.

Kurvlängden betecknas: S

Större bild av bågelement



Pythagoras sats ger
$(ds)^2 = (dx)^2 + (dy)^2$
- ds är båglängd, > 0
- $ds = \sqrt{(dx)^2 + (dy)^2}$

Integrationen blir då:

$$\int ds = \int_a^b \sqrt{(dx)^2 + (dy)^2}$$

Vi får då att kurvlängden ges av

$$\boxed{S = \int_a^b \sqrt{(dx)^2 + (dy)^2}} \qquad a \le x \le b$$

Vad vi integrerar med avseende på får vi utifrån frågan i uppgiften.

Ex: Bestäm längden av kurvan där

$f(x) = \frac{x^2}{4} - \frac{\ln x}{2}, \quad 1 \leq x \leq 4$

Lösning:

• Börja med att utgå från:

$$S = \int_1^4 \sqrt{(dx)^2 + (dy)^2} = \int_1^4 \sqrt{\left(\frac{(dx)^2}{(dx)^2} + \frac{(dy)^2}{(dx)^2}\right) \cdot (dx)^2}$$

$$= \int_1^4 \sqrt{1 + \frac{(dy)^2}{(dx)^2}} \cdot \sqrt{(dx)^2} = \left[\begin{array}{l} \sqrt{(dx)^2} = |dx| \\ dx > 0 \text{ så} \\ \sqrt{(dx)^2} = |dx| = dx \end{array}\right]$$

$$= \int_1^4 \sqrt{1 + (f'(x))^2}\, dx$$

$$= \left[f(x) = \frac{x^2}{4} - \frac{\ln x}{2} \right]$$

$$= \int_1^4 \sqrt{1 + \left(\frac{1}{4} \cdot 2x - \frac{1}{2} \cdot \frac{1}{x}\right)^2} dx = \int_1^4 \sqrt{1 + \left(\frac{1}{2}x - \frac{1}{2x}\right)^2} dx$$

$$= \int_1^4 \sqrt{1 + \left(\frac{1}{2}\left(x - \frac{1}{x}\right)\right)^2} dx = \int_1^4 \sqrt{1 + \frac{1}{4}\left(\frac{x^2 - 1}{x}\right)^2} dx$$

$$\int_1^4 \sqrt{1 + \frac{1}{4}\left(\frac{x^4 - 2x^2 + 1}{x^2}\right)} dx$$

$$= \int_1^4 \sqrt{1 + \frac{x^4 - 2x^2 + 1}{4x^2}} dx = \int_1^4 \sqrt{\frac{4x^2}{4x^2} + \frac{x^4 - 2x^2 + 1}{4x^2}} dx$$

$$= \int_1^4 \sqrt{\frac{x^4 + 4x^2 - 2x^2 + 1}{4x^2}} dx = \int_1^4 \sqrt{\frac{x^4 + 2x^2 + 1}{4x^2}} dx$$

$$= \int_1^4 \frac{\sqrt{(x^2+1)^2}}{\sqrt{4x^2}} dx = \int_1^4 \frac{|x^2+1|}{2|x|} dx = \left(\begin{array}{l} |x^2+1| = x^2+1 \\ \text{eftersom } x^2+1 \geq 1 \\ |x| = x \text{ i } x > 0 \\ \text{eftersom } 1 \leq x \leq 4 \end{array}\right)$$

$$= \int_1^4 \left(\frac{x^2+1}{2x}\right) dx = \int_1^4 \left(\frac{x^2}{2x} + \frac{1}{2x}\right) dx$$

$$= \int_1^4 \left(\frac{1}{2}x + \frac{1}{2x}\right) dx \quad \frac{1}{2}\int \left(x + \frac{1}{x}\right) dx$$

$$= \frac{1}{2}\left[\frac{x^2}{2} + \ln|x|\right]_1^4$$

$$= \frac{1}{2}\left(\left(\frac{4^2}{2} + \ln|4|\right) - \left(\frac{1^2}{2} + \ln|1|\right)\right)$$

$$= \frac{1}{2}\left(\left(\frac{1}{2}\cdot 16 + \ln|4|\right) - \left(\frac{1}{2} + \underbrace{\ln|1|}_{=0}\right)\right)$$

$$= \frac{1}{2}\left(\left(8 + \ln|4|\right) - \left(\frac{1}{2}\right)\right)$$

$$= \frac{1}{2}\left(8 - \frac{1}{2} + \ln|4|\right) = \frac{1}{2}\left(\frac{16}{2} - \frac{1}{2} + \ln|4|\right)$$

$$= \frac{1}{2}\left(\frac{15}{2} + \ln|4|\right) = \frac{1}{2}\left(\frac{15}{2} + \ln(2^2)\right)$$

$$= \frac{1}{2}\left(\frac{15}{2} + 2\ln(2)\right) = \frac{15}{4} + \ln(2)$$

<u>**SVAR:**</u> $\frac{15}{4} + \ln(2)$ l e

Ex: Beräkna längden av kurvan

$$\begin{cases} x = e^{-t}\cos(t) \\ y = e^{-t}\sin(t) \end{cases}, \quad 0 \le t \le 2\pi$$

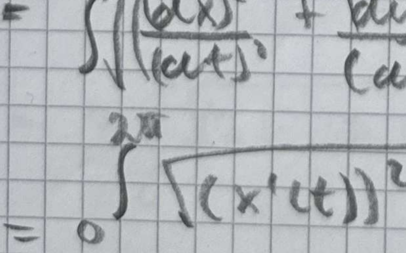
OBSERVERA:

$x = x(t)$

$y = y(t)$

Man behöver ej rita ytan det görs för förståelse.

t	$x = e^{-t}\cos(t)$	$y = e^{-t}\sin(t)$	(x,y)
0	$e^0 \cdot \cos(0) = 1$	$e^0 \sin(0)$	$(1,0)$
$\pi/2$	$e^{-\pi/2}\cos(\pi/2) = 0$	$e^{-\pi/2}\sin(\pi/2) = \frac{1}{e^{\pi/2}}$	$(0, 1/e^{\pi/2})$
π	$e^{-\pi}\cos(\pi) = -\frac{1}{e^{\pi}}$	$e^{-\pi}\sin(\pi) = 0$	$(-\frac{1}{e^{\pi}}, 0)$
2π	$e^{-2\pi}\cos(2\pi) = \frac{1}{e^{2\pi}}$	$e^{-2\pi}\sin(2\pi) = 0$	$(\frac{1}{e^{2\pi}}, 0)$



$$ds = \sqrt{(dx)^2 + (dy)^2} \Rightarrow$$

$$S = \int \sqrt{(dx)^2 + (dy)^2} =$$

$$= \int \sqrt{\left(\frac{(dx)^2}{(dt)^2} + \frac{(dy)^2}{(dt)^2}\right)(dt)^2} = \left[\frac{(dx)^2}{(dt)^2} = \left(\frac{dx}{dt}\right)^2 = (x'(t))^2\right]$$

$$= \int_0^{2\pi} \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$$

$$= \int_0^{2\pi} \sqrt{\left(-e^{-t}\cos(t) + e^{-t}(-\sin(t))\right)^2 + \left(-e^{-t}\sin(t) + e^{-t}\cos(t)\right)^2}\, dt$$

$$= \int_0^{2\pi} \sqrt{(e^{-t})^2\left(\left(-\cos(t)-\sin(t)\right)^2+\left(-\sin(t)+\cos(t)\right)^2\right)}\,dt$$

$$= \int_0^{2\pi} \sqrt{(e^{-t})^2}\sqrt{\left(\cos(t)+\sin(t)\right)^2+\left(\cos(t)-\sin(t)\right)^2}\,dt$$

$$= \int_0^{2\pi} e^{-t}\sqrt{\underbrace{\cos^2(t)+\sin^2(t)}_{\text{trig ettan}}+\cancel{2\sin(t)\cos(t)}} \ldots$$

$$\ldots \sqrt{\underbrace{\cos^2(t) - \cancel{2\cos(t)\sin(t)} + \sin^2(t)}_{\text{trig ettan}}}\,dt$$

$$= \int_0^{2\pi} e^{-t}\cdot\sqrt{2}\,dt = \sqrt{2}\int_0^{2\pi} e^{-t}\,dt =$$

$$= \sqrt{2}\left[\frac{e^{-t}}{-1}\right]_0^{2\pi} = -\sqrt{2}\left[e^{-t}\right]_0^{2\pi} =$$

$$= -\sqrt{2}\left(e^{-2\pi}-e^{0}\right) = -\sqrt{2}\left(e^{-2\pi}-1\right)$$

$$= \sqrt{2}\left(1-e^{-2\pi}\right)$$

SVAR:

$$S = \sqrt{2}\left(1-e^{-2\pi}\right) \text{ l.e}$$

Kurvor på POLÄRFORM

Ex:

Beräkna längden av $r = 1 + \cos\varphi$, $-\pi \leq \varphi \leq \pi$

Glöm ej att

$$\begin{cases} x = r\cos\varphi = \\ y = r\sin\varphi = \end{cases}$$

$$ds = \sqrt{\left(\frac{(dx)^2}{(d\varphi)^2} + \frac{(dy)^2}{(d\varphi)^2}\right)(d\varphi)^2} =$$

$$= \sqrt{(x'(\varphi))^2 + (y'(\varphi))^2}\, d\varphi$$

OBS: Finns även formel som kan nyttjas:

$$ds = \sqrt{r^2 + (r')^2}\, d\varphi$$

$$s = \int_{-\pi}^{\pi} \sqrt{(r(\varphi))^2 + (r'(\varphi))^2}\, d\varphi$$

$$= 2\int_0^{\pi} \sqrt{(1+\cos(\varphi))^2 + (-\sin(\varphi))^2}\, d\varphi$$

$$= 2\int_0^{\pi} \sqrt{1 + 2\cos(\varphi) + \underbrace{\cos^2(\varphi) + \sin^2(\varphi)}_{\text{trig ettan}}}\, d\varphi$$

$$= 2\int_0^{\pi} \sqrt{2 + 2\cos(\varphi)}\, d\varphi = 2\sqrt{2}\int_0^{\pi} \sqrt{1 + \cos(\varphi)}\, d\varphi$$

= [Nyttja Dubbelvinkel
$\cos(2x) = 2\cos^2(x) - 1$
$\cos(\varphi) + 1 = 2\cos^2\left(\frac{\varphi}{2}\right)$
| $\cos(2x) + 1 = 2\cos^2(x)$
Sätt $2x = \varphi$
$x = \varphi/2$]

$$= 2\sqrt{2}\int_0^{\pi} \sqrt{2\cos^2\left(\frac{\varphi}{2}\right)}\, d\varphi =$$

[dra ut $\sqrt{2}$!]

$$= 4\int_0^{\pi} \sqrt{\cos^2\left(\frac{\varphi}{2}\right)}\, d\varphi = 4\int_0^{\pi} \left|\cos\left(\frac{\varphi}{2}\right)\right| d\varphi$$

$$= \left[\left|\cos\left(\frac{\varphi}{2}\right)\right| = \cos\left(\frac{\varphi}{2}\right) \text{ då } 0 \leq \varphi \leq \pi \right]$$

$$= 4\int_0^{\pi} \cos\left(\frac{\varphi}{2}\right) d\varphi = \left[\begin{array}{l} \frac{\varphi}{2} = t \\ \varphi = 2t \\ d\varphi = 2\,dt \\ \text{nya gränser} \\ \hline \varphi = 0 \Rightarrow t = 0 \\ \varphi = \pi \Rightarrow t = \frac{\pi}{2} \end{array} \right]$$

$$= 4\int_0^{\pi/2} \cos(t)\, 2\,dt = 8\int \cos(t)\, dt$$

$$= 8\left[\sin(t)\right]_0^{\pi/2} = 8\left(\sin\left(\frac{\pi}{2}\right) - \sin(0)\right)$$

$$= 8(1-0) = 8$$

<u>SVAR:</u> 8. l.e

SAMMANFATTNING

$$s = \int_x^y \sqrt{1 + (f'(x))^2}\, dx$$

$$s = \int_{t_1}^{t_2} \sqrt{(x'(t))^2 + (y'(t))^2}\, dt$$

$$s = \int_\alpha^\beta \sqrt{(r(\varphi))^2 + (r'(\varphi))^2}\, d\varphi$$

Alla dessa härleds från:

$$ds = \sqrt{(dx)^2 + (dy)^2}$$

EXTRA VARIABELBYTE

$$\cos(2\varphi) = 2\cos^2(\varphi) - 1$$

$$\cos(2\varphi) + 1 = 2\cos^2(\varphi)$$

$$\boxed{\text{Sätt } 2\varphi = x}$$

$$\boxed{\cos(x) + 1 = 2\cos^2\left(\frac{x}{2}\right)}$$

ELLER

$$\cos(2\varphi) = 1 - \sin^2(\varphi)$$

$$1 - \cos(2\varphi) = 2\sin^2(\varphi)$$

Sätt $2\varphi = x$

$$\boxed{1 - \cos(x) = 2\sin^2\left(\frac{x}{2}\right)}$$